

Stanley Chow

pingsenchow@outlook.com | +852 5731 5812 | <https://github.com/Stanley-Chow>

EDUCATION & HONORS

The University of Hong Kong

Bachelor of Science in Mathematics and Computer Science

GPA: 3.98/4.30 | **Honors:** Dean's List (2025), UG Entrance Scholarship, Lee Shau Kee Scholarship

Mathematics Coursework: Probability, Stochastic Process, Optimization, Numerical Analysis, Scientific Computing

Computer Science Coursework: Algorithm Design and Analysis, Data Structures, Intro to Data Science, Machine Learning

Hong Kong

May 2028

TECHNICAL SKILLS

Languages: Python, C++, MATLAB, SQL. | **Tools/Utilities:** Git/GitHub, LaTeX

ML Frameworks: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, SciPy, PyTorch, LightGBM, Transformers, RAG.

Quantitative Methods: Factor Modeling, Statistical Learning, Numerical Optimization, Monte Carlo Simulation, PDE Solvers.

WORK EXPERIENCE

Faculty of Science, University of Hong Kong

Incoming HKU Summer Research Fellow

Hong Kong

May 2026 – Present

- Develop a data-driven POD-based model to reconstruct time-dependent heat sources with uncertain data (parabolic inverse source problems); reduce a high-dimensional problem ($O(10^4)$ degree of freedom) to ~ 20 -dimensional solution basis.
- Design a posteriori algorithm to find the optimal regularization parameters in the optimization problem using the proposed POD method without the noise-level knowledge, reducing the relative reconstruction error by 26%.
- Conduct stochastic error estimation using Monte Carlo simulations on 100+ test cases to quantify model robustness and demonstrate the model's computational acceleration ($6.5\times$ speedup, $1328s \rightarrow 164s$) compared to full-order methods.

Wuhan Boeyo Disaster Reduction Technology Co., Ltd. | Leading Seismic Instrumentation Company

Data Engineer Intern

Wuhan, China

Jun 2024 – Sep 2024

- Engineered Python-based validation pipelines for BY-series EEW terminals, with 2000+ units deployed, that integrated global real-time seismic data with local sensor inputs to achieve full validation coverage for life-safety alerts.
- Developed signal processing algorithms, including STA/LTA event detection, Weighted Voting Mechanism, and Dynamic Event Timing Logic, lowering false positive rate by 15% ($21 \rightarrow 17$ alerts per day) in high-precision safety alerts.
- Architected real-time visualization dashboards demo using Matplotlib and Seaborn to translate seismic waveforms and terminal performance metrics for 5+ potential clients in the school, gas, and rail sectors.

SELECTED RESEARCH

Neural-Augmented Reduced-Order Methods for PDE Inverse Problems | *In Preparation*

Jun 2026 – Present

Urban Transportation Network Optimization Research | *S.-T. Yau High School Science Award*

May 2023 – Nov 2023

- Developed mathematical optimization models for urban rail transit design using geometric constraints and construction cost minimization objectives; derived analytical solutions for single-intersection routing cases using linear programming methods.
- Proposed multi-intersection track optimization algorithms using Particle Swarm Optimization (PSO), reducing simulated route length and construction cost while satisfying curvature and engineering feasibility constraints.

SELECTED PROJECTS

WorldQuant International Quant Championship (IQC)

Apr 2026 – May 2026

- Conducted quantitative factor research on the WorldQuant Brain platform using 5+ years of US equity market data; developed and backtested 40+ alpha factors across momentum, value, volatility, and sentiment dimensions, achieving an average Sharpe Ratio of 3.55 through iterative strategy optimization and Long-Short portfolio construction.
- Built a multi-factor stock selection framework under IQC evaluation standards by applying factor combination, normalization, and risk control techniques; optimized portfolio robustness and turnover efficiency, resulting in a hypothetical margin of 14% and ranking within Top 5 in Hong Kong among 1,500+ participants (Gold Medal in WorldQuant Challenge).

DocuQuest Agent | [Open-source](#) | [Demo](#)

Feb 2026 – Mar 2026

- Developed a Retrieval-Augmented Generation (RAG) pipeline in C++, including tokenization, inverted index construction, and LLM-guided context synthesis, to enable context-based inquiry across 40+ unstructured text files.
- Implemented an LLM-guided Query Expansion strategy using two-stage prompting to transform natural language questions into search terms, increasing the retrieval hit rate by $\sim 20\%$ compared to baseline keyword matching.
- Optimized the retrieval-to-generation bottleneck by implementing an $O(\log N + V)$ Inverted Index, reducing $\sim 10\%$ latency.